

Assignment

Tuesday, 10 August 2021

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Q1

Show that if two events A and B are independent, then A^c and B^c are also independent.

Soln

If A and B are independent events, then

$$P(A \cap B) = P(A)P(B) \quad \dots (i)$$

To show A^c and B^c are also independent

$$P(A^c \cap B^c) = P(A^c)P(B^c)$$

$$P(A^c \cap B^c) = P(\overline{A \cup B}) \quad \text{Using De Morgan's Law}$$

$$= 1 - P(A \cup B)$$

$$= 1 - [P(A) + P(B) - P(A \cap B)]$$

$$= 1 - [P(A) + P(B) - P(A)P(B)] \quad \text{(As we know A and B are independent)}$$

$$= 1 - [1 - P(A^c) + 1 - P(B^c) - (1 - P(A^c))(1 - P(B^c))] \quad \text{Using (i)}$$

$$= 1 - [1 - P(A^c) + 1 - P(B^c) - 1 + P(A^c) + P(B^c) - P(A^c)P(B^c)]$$

$$= 1 - 1 + P(A^c) + P(B^c) - P(A^c)P(B^c)$$

$$= P(A^c)P(B^c)$$

$$\text{We proved } P(A^c \cap B^c) = P(A^c)P(B^c)$$

Hence, A^c and B^c are also independent

As we know if two events X and Y are independent, then

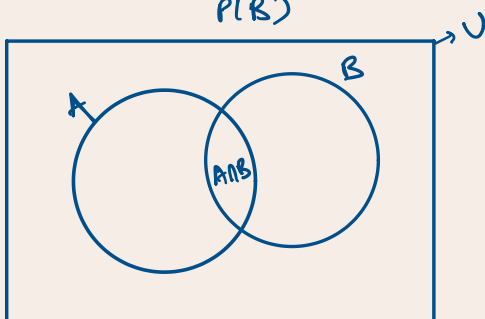
$$P(X \cap Y) = P(X)P(Y)$$

Q2

Soln

Assume that $P(B) > 0$. Show that $P(A|B) = 1 - P(A^c|B)$.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



$$P(A^c \cap B) = P(B) - P(A \cap B)$$

$$\Rightarrow P(A \cap B) = P(B) - P(A^c \cap B) \quad \dots (i)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{P(B) - P(A^c \cap B)}{P(B)} \quad \dots \text{Using (i)}$$

$$= 1 - \frac{P(A^c \cap B)}{P(B)}$$

$$= 1 - P(A^c|B) \quad \text{Hence, Proved}$$

Q3

A health maintenance organization (HMO) concerned with the adverse financial effects of catastrophic medical expenses for its members enters into a reinsurance arrangement in which the reinsurer (i.e., the insurer providing insurance to the HMO) agrees to pay the amount of each plan member's annual hospital expenses that exceed \$100,000. Thus, for example, if the HMO had exactly two members and these two members had annual expenses of \$130,000 and \$70,000, respectively, then the reinsurer would pay the HMO \$30,000. Under the terms of the agreement, the HMO will be required to pay additional reinsurance premium if the fraction of the plan membership for which the reinsurance claim is filled exceeds 5%. This condition helps minimize any incentive the HMO may have to increase its enrolment of substandard risks after entering into the reinsurance arrangement.

The probability of catastrophic claim for any given member clearly depends on many factors, one of which is the age of the member. On the basis of historical data, the plan estimates that 10% of members over the age of 65 will incur hospital expenses exceeding \$100,000, 4% of members between the ages of 40 and 65 will incur hospital expenses exceeding \$100,000, and 3% of members under the age of 40 will incur hospital expenses exceeding \$100,000.

Twenty percent of the HMO's members are currently over the age of 65. If the HMO wishes to maintain this level of members over the age of 65 (perhaps for competitive or regulatory reasons) and wishes to avoid exceeding the 5% trigger point in the reinsurance agreement, how large must the HMO's under-40 population be?

Let's say there are total of 100 people with expenses $> 100K$

$$N_1 = \text{Number of people } (< 40) \Rightarrow N_1 + N_2 + N_3 = 100$$

$$N_2 = \text{Number of people } (40-65)$$

$$N_3 = \text{Number of people } (> 65)$$

$$\text{At } N_3 (> 100K) = 0.1 N_3 \quad \text{Also } N_3 = 20$$

$$N_2 (> 100K) = 0.04 N_2 \quad \text{Let's say } N_1 = X$$

$$N_1 (> 100K) = 0.03 N_1 \quad \text{then } N_2 = 100 - N_1 - N_3$$

$$\text{Total No. of people } (> 100K) = 0.03 N_1 + 0.04 N_2 + 0.1 N_3 = \frac{100 - X - 20}{80 - X}$$

For reinsurance not to occur, the total number of people with $> 100K$ of Medical Expenses should be $< 5\%$ i.e. < 5

$$\Rightarrow 0.03 N_1 + 0.04 N_2 + 0.1 N_3 < 5$$

$$\Rightarrow 0.03 X + 0.04(80 - X) + 2 < 5$$

$$\Rightarrow 0.01 X > 2$$

$$\Rightarrow \underline{X > 20} \quad \boxed{X > 20\%} \quad \underline{\text{Ans}}$$

Q4

A box contains three coins. Two of these are fairly unusual coins: one has heads on both sides, one has tails on both sides. The other is a fair coin.

One of the three coins is chosen randomly (with each coin equally likely) and then it is tossed

once. The toss comes up heads. What is the probability that the coin chosen is the fair coin?

Soln

We have total 6 probable outcomes.

$$\left. \begin{array}{l} H_1 \text{ of } H_1 H_1 \text{ coin} \\ H_2 \text{ of } H_1 H_1 \text{ coin} \\ H \text{ of } HT \text{ coin} \end{array} \right\} 3 \text{ Heads} \rightarrow P(H|H) = \frac{1}{3}$$

$$\left. \begin{array}{l} T \text{ of } HT \text{ coin} \\ T_1 \text{ of } T_1 T_1 \text{ coin} \\ T_2 \text{ of } T_1 T_1 \text{ coin} \end{array} \right\} 3 \text{ Tails}$$

We have already given that the toss came up as Heads

$$P(H|H) = \frac{P(H|H_1)P(H_1) + P(H|H_2)P(H_2) + P(H|H_3)P(H_3)}{P(H|H_1)P(H_1) + P(H|H_2)P(H_2) + P(H|H_3)P(H_3)}$$

$$= \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2} \times \frac{1}{3} + 1 \times \frac{1}{3} + 0}$$

$$= \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{3} \times \frac{2}{3}} = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3} \quad \underline{\text{Ans}}$$

Q5

Suppose that events A_1 and A_2 are independent conditional on event B .

Show that $P(A_2 | A_1 \cap B) = P(A_2 | B)$. You can assume that $P(A_1 \cap B) > 0$.

Soln

$$P(A_1|B)P(A_2|B) = \frac{P(A_1 \cap A_2)}{P(B)}$$

$$P(A_2|B) = \frac{P(A_1 \cap A_2)}{P(A_1 \cap B)}$$

$$P(A_2|B) = \frac{P(A_2 \cap B)}{P(B)}$$

$$P(A_1 \cap A_2|B) = \frac{P(A_1 \cap A_2 \cap B)}{P(B)}$$

$$\frac{P(A_1 \cap B)}{P(B)} \times \frac{P(A_2|B)}{P(B)} = \frac{P(A_1 \cap A_2 \cap B)}{P(B)}$$

$$\frac{P(A_1 \cap B)}{P(B)} = \frac{P(A_1 \cap A_2 \cap B)}{P(A_2 \cap B)}$$

$$P(A_2|B) = \frac{P(A_1 \cap A_2 \cap B)}{P(A_1 \cap B)} \quad \underline{\text{Ans}}$$

Solved